Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/01/2024
PI Name:	Alwood, Joshua Ph.D.		
Project Title:	DAG Studies for the Behavioral Medicine Risk		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-1490
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Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	06/01/2023	**End Date:**	03/31/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2025 per L. Juliette/NASA HRP (Ed., 9/30/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Antonsen, Eric M.D., Ph.D. (Potomac Institute) Reynolds, Robert Ph.D. (KBR Wyle) Dev, Sheena Ph.D. (KBR Wyle) Nelson, Gregory (KBR Wyle)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Determine how we can use empirical data sets from animal experiments after exposure to simulated or actual spaceflight hazards to test the veracity of the Behavioral Medicine (Bmed) Directed Acyclic Graph (DAG). Published quantitative analytics (including DAG and specific-DAG-pathway Quality of Evidence and Level of Evidence scoring approaches) will be assessed for applicability to radiation risk analyses. Aim: Develop and validate portions of the Behavioral Medicine Directed Acyclic Graph (DAG) using animal data. Goals for this task include: 1. Evaluate if and how animal data support the DAG method for the Bmed risk assessment. 2. Evaluate if and how animal data informs human Bmed risk assessment. 3. Evaluate if and how radiation animal literature informs both risk assessment, interventions, and thresholds of permissible outcome limits for Bmed human performance. 4. Evaluate if and how radiation-induced rodent effects translate to human data and inform radiation countermeasure recommendations. 5. Identify if and how scaling factors might be applied to rodent data in order to inform Bmed human risk assessment.
Rationale for HRP Directed Research:	This task is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	This work strives to demonstrate/assess the translational potential of animal studies for risk management for humans, with applicability for humans on Earth and in space.
Task Progress:	Over the past 9 months, our team has developed and demonstrated a workflow for Directed Acyclic Graph (DAG) analyses using raw data available in the public domain or from PI correspondence. We presented our first complete analysis at the 2024 NASA's Human Research Program (HRP) Investigators' Workshop. To achieve a complete preliminary analysis, we worked with data available on the NASA Open Science Data Repository (OSDR) or direct PI correspondence. These 4 studies were assessed for Quality of Evidence below, data linked with specific DAG nodes, and assessed for conditional independence and marginal correlations using DAGitty in R. This first analysis will allow for fine-tuning of the workflow for subsequent larger analyses.
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Abstracts for Journals and Proceedings	Alwood J, Antonsen E, Dev S, Nelson G, Reynolds R, Dvorochkin T, Shahid A. "DAG studies for the behavioral medicine risk." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024